

Mathematics

Stage 9

Paper 1

2025

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Square brackets in mark scheme

When square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6[58...]	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30..... cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

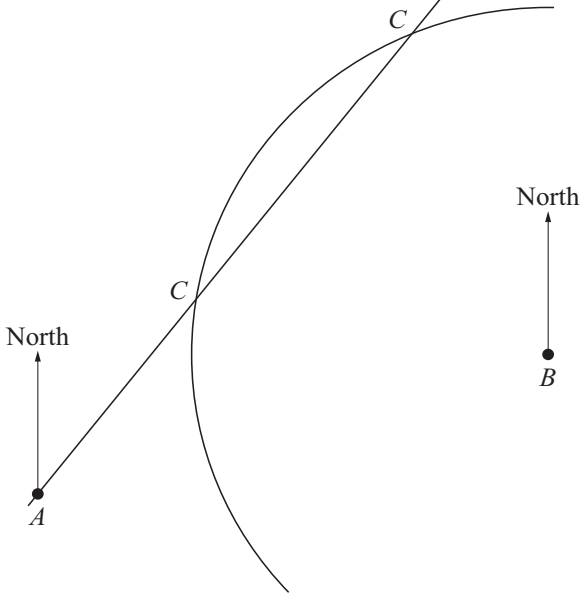
Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance						
1	(Largest $\Rightarrow$) 3.2×10^6 (Smallest $\Rightarrow$) 5.0×10^{-8}	2	Award 1 mark for each correct answer.							
2	128 GB	1		Accept any clear indication.						
3	Any two pairs of numbers such that $p + q = 2$ e.g. two from ($p =$) 1 and ($q =$) 1 ($p =$) 0 and ($q =$) 2 ($p =$) 2 and ($q =$) 0	2	Award 1 mark for each correct pair.	Accept fractional, decimal or negative numbers, e.g. ($p =$) 0.5 and ($q =$) 1.5 ($p =$) 3 and ($q =$) -1						
4(a)	148	1								
4(b)	$n^2 + 6$	1		Accept equivalent answers, e.g. $n^2 + 4 + 2$						
5	$\sqrt{5}$ and π	1		Both answers correct for the mark. Accept any clear indication.						
6(a)	Two correct entries in the table. <table><tr><td>(A)</td><td>(120°)</td><td>60(°)</td></tr><tr><td>(B)</td><td>108(°)</td><td>(72°)</td></tr></table>	(A)	(120°)	60(°)	(B)	108(°)	(72°)	1		Both answers correct in the correct spaces for the mark.
(A)	(120°)	60(°)								
(B)	108(°)	(72°)								
6(b)	The interior angle [of polygon A] is a factor of 360°.	1		Accept any equivalent answer, e.g. 120° divides exactly into 360°.						

Question	Answer	Marks	Part Marks	Guidance						
7	$(b =) \ [\pm] \ \sqrt{a + 4}$	2	Award 1 mark for correct first step $a + 4 = b^2$ or for correct answer followed by incorrect subsequent working.							
8	(0, 3)	1		Accept any clear indication.						
9	<table border="1"><tr><td>True</td><td>False</td></tr><tr><td>✓</td><td></td></tr><tr><td></td><td>✓</td></tr></table>	True	False	✓			✓	1		Both answers correct for the mark. Accept any clear indication.
True	False									
✓										
	✓									
10(a)	Five correct probabilities correctly placed on tree diagram $\frac{3}{4} \ , \ \frac{2}{5} \ , \ \frac{3}{5} \ , \ \frac{2}{5} \ , \ \frac{3}{5}$	2	Award 1 mark for at least one correct probability correctly placed.	Accept equivalent fractions, decimals or percentages.						
10(b)	$\frac{9}{20}$	2	Award 1 mark for $\frac{3}{4} \times \frac{3}{5}$ or for <i>their</i> $\frac{3}{4} \times$ <i>their</i> $\frac{3}{5}$	Accept equivalent fractions, decimals or percentages. For 1 mark, <i>their</i> $\frac{3}{4}$ and <i>their</i> $\frac{3}{5}$ must be between 0 and 1						

Question	Answer	Marks	Part Marks	Guidance
11(a)	30(cm)	1		
11(b)	44(cm ²)	1		
12	Any correct fraction in the given range, e.g. $\frac{11}{26}$	1		Numerator and denominator must be integers.
13(a)	0.35	1		
13(b)	12	1		
13(c)	20	1		
14	314 cm ³	1		Accept any clear indication.
15	Output of 1.25 Inputs of 4 and -4	2	Award 1 mark for one or two correct answers.	Accept equivalent values for 1.25, e.g. $\frac{5}{4}$ Accept inputs of 4 and -4 in either order.
16	Hexagon	1		

Question	Answer	Marks	Part Marks	Guidance
17	$(x =) 2.5$ $(y =) -1$	3	Award 2 marks for one correct value or Award 1 mark for a correct method to eliminate one variable or for <i>their</i> two values satisfying one of the original equations.	Accept equivalent values for 2.5, e.g. $\frac{5}{2}$ Correct method, e.g. Multiplying one or both equations by a constant to get equal coefficients and correct consistent addition or subtraction or making x or y the subject and substituting into the other equation.
18(a)	$(x =) 0.8[0]$	1		
18(b)	(\$)17.50	1		

Question	Answer	Marks	Part Marks	Guidance
19	The two possible positions of C marked 6 cm from B and on a bearing of 040° from A. 	3	Award 2 marks for C marked 6 cm from B. or Award 1 mark for C marked on a bearing of 040° from A and Award 1 mark for $300\,000 \div 50\,000$ seen or implied.	2 marks implied by one correct position of C. 1 mark implied by 6 cm seen.
20	4 -3	2	Award 1 mark for each correct answer.	
21	$4xy + 1$ (cm)	2	Award 1 mark for $4x(4xy + 1)$ or for either $4xy$ or 1 correct in a two-term expression.	
22	rotation and 180° and [centre of rotation] $(1, -2)$	2	Award 1 mark for two correct from - rotation 180° [centre of rotation] $(1, -2)$	Award 2 marks for enlargement s.f. -1 about $(1, -2)$. Award 1 mark for enlargement s.f. -1 Award 0 marks for more than one transformation.

Question	Answer	Marks	Part Marks	Guidance
23	$1\frac{17}{20}$	3	Award 2 marks for an answer equivalent to $1\frac{17}{20}$ or Award 1 mark for $\frac{14}{15} \times \frac{3}{8} = \frac{7}{20}$ or equivalent or for $2\frac{1}{5} = \frac{11}{5}$ or for correctly changing <i>their</i> $\frac{11}{5}$ and $\frac{7}{20}$ to a common denominators prior to subtraction If 0 marks scored, award 1 mark for answer $\frac{19}{40}$ (incorrect order of operations).	For 2 marks, e.g. $\frac{37}{20}$, $1\frac{85}{100}$, 1.85, ...

Question	Answer	Marks	Part Marks	Guidance
24	$n = 10$ shown in the working and (Answer =) 11	4	Award 3 marks for $n = 10$ or Award 1 mark for $\frac{n+12}{2} = \frac{3}{2}n - 4$ or equivalent and Award 1 mark for correct first step in solving <i>their</i> equation in one unknown with unknown on both sides.	Accept any letter for the unknown. Accept 10 on the answer line for 3 marks. Or equivalent, e.g. $(x + 12) \div 2 = 3 \times x \div 2 - 4$ If from correct equation, correct first step could be, e.g. $n + 12 = 3n - 8$ or $\frac{n}{2} + 6 = \frac{3n}{2} - 4$
25	Any value for p in range $-10 < p < -5$	1		

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